Photosynthesis State-Machine Collage

Report



Overview

Photosynthesis State-Machine Collage draws inspiration from **the Rhythms of Photosynthesis**, **Collaged imagery**, **Non-linear Ambient generative music**, **State-machine structure**, and interactive installation elements side by side to form a slowly evolving interactive audio-visual system.

The work begins with the textures of leaves and their interaction with shifting light, breaking them apart, recomposing and collaging them. At the same time, different “states” collectively shape the direction of the overall sound, allowing the collage system to unfold like a garden composed of plants that continuously regulate themselves in response to the light by **Audience interaction**.

The sonic layer centres on gradients, repetition, and fluid motion. Through algorithms and real-time mapping, it builds a breathing-like pulse across multiple voices. The visual growth and the movement of sound pull on each other, forming an ecology that sits between the natural and the mechanical.

Through real-time interaction, the viewers encounter not a simple representation of photosynthesis, but a process of absorption, transformation, accumulation, and growth—an audio-visual apparatus that thinks and behaves in plant-like way.

Aim & Aesthetic

Following an earlier piece, I wanted to expand my practice into a broader set of technical directions while further developing my aesthetic sensibilities. For this piece, I therefore decided to work with both the Arduino-based interaction techniques I had covered earlier and my initial explorations with trained models for nn~. Although the neural-model experiments were not ultimately successful in the final patch, they informed my understanding of machine-learning-driven sound processing and will be mentioned later in this report reflection part.

At the same time, I wanted to explore collage-based visual methods. My strong affinity for the natural world also played an important role. One day, while walking across a thick layer of fallen leaves in a park—hearing the soft rustling underfoot and seeing the mixture of yellow and green tones—I suddenly imagined an ambient soundscape that could accompany this experience. This moment became the core inspiration for Photosynthesis State-Machine Collage.

Brian Eno's work has also had a significant influence on this project, such as *Making Gardens Out of Silence*—structures that grow, transform, and respond over time. Furthermore, Eno's interview *Answers Deep Questions on Creativity, Generative Music & Future of Sound* *https://youtu.be/JzX7pLKCnMw?si=idHDY_-aebKRhODq* offered valuable insights into the role of unpredictability, constraint, and creative mapping in generative music. These ideas directly shaped the conceptual and aesthetic foundations of my project, particularly in how I approached states of growth, sonic evolution, and the overall design of an interactive system.

Photosynthesis State-Machine Collage User Guide

Browse the file "USER_GUIDE.docx" in this repository.

System Architecture

The entire work consists of four parts: External Interaction,State-Machine, Sound System and Visual Engine.

Interaction (Arduino)

In this project, I incorporated Arduino-based external interaction as a core component of the system. I used four light sensors to capture real-time illumination data, enabling the installation to respond dynamically to changes in light intensity and position. To support this, I generated and refined the necessary Arduino code with the assistance of AI which allowed me to efficiently prototype functions such as calculating the coordinates of the light-spot centre and extracting stable brightness values. The code is included in the submitted project folder.

This interactive sensing layer forms the primary technical foundation for the aesthetic mapping in the work. It is also my first attempt at integrating physical interaction into a formal project, marking a significant expansion of my technical practice and shaping the way the sonic and visual behaviours unfold within the piece.

State-Machine

I designed a state-machine system grounded in an aesthetic mapping inspired by plant photosynthesis. The system begins with a simple rule: exposure to light causes a plant unit to brighten (in terms of visual luminance) and then to "grow." Growth is expressed both visually—through a colour transition from yellow to green—and sonically, through corresponding changes in the behaviour of each voice within the composition. Conversely, when a unit receives no light or moves away from the illuminated region, it begins to decay either immediately or after a delay, depending on whether it has reached its "mature" stage.

To achieve this, I implemented a set of real-time calculations that assign each unit a dynamic light-intensity variable (*intensity_leaf_n*). This value is then processed through growth and decay coefficients, producing a continuously updated growth parameter (*leaf_n_GrowthState*). The growth state drives multiple layers of the system:

the colour progression and visual presence of each plant unit,

the rhythmic and behavioural pacing of its associated sound voice, and

the accumulation count of mature units within the overall system.

Once the number of matured units reaches a defined global threshold, the state machine iterates to the next system-wide state. This creates a feedback loop in which local interactions (individual leaves responding to light) collaboratively influence global behaviour across the entire audiovisual ecosystem.

Sound System

Timbre

The sound part is composed of three types: note type, drone and ambient environment sound.

Within Max/MSP, I designed several note-like timbres using waveform synthesis and modulation techniques. In addition to these internally generated voices, I also loaded several piano-based sound sources through the VST~ object, forming the majority of the harmonic and melodic layers in the system. Each voice is then processed individually through Valhalla reverb instances, with different parameter settings applied to create depth, spatial variation, and a sense of organic motion.

The lotus form at the centre of the visual field functions as the trigger unit for a drone-based harmonic layer. I constructed this drone using multiple Cycle~ oscillators, combined to form a sustained chord. And LFO continuously modulates aspects of the timbre, producing a subtle, non-linear fluctuation. This behaviour is further influenced by the viewer's illuminated position on screen, creating a dynamic relationship between light input and the perceived harmonic stability of the drone.

As described earlier, once the number of units reaching the "mature" stage exceeds specific thresholds, the system also triggers two ambient environmental layers: a soft leaf-rustling texture and intermittent bird calls. These layers reinforce the ecological metaphor of the work and expand the sonic environment beyond the purely synthesized voices.

composition

In terms of MIDI parameters, the global harmonic framework is based primarily on Cmaj7 and Fmaj7, which function as the core tonal centres of the system. The transitions between these two harmonies are determined by the number of units that have entered the *mature* stage. When the system reaches the highest maturity threshold, a global key shift is triggered, allowing the harmonic field to iterate and evolve. This mechanism establishes a dynamic harmonic structure in which the tonal environment responds directly to the progression of the state-machine.

In terms of compositional structure, each state within the state-machine is assigned its own algorithmic logic, combining deterministic rules with controlled randomisation for pattern generation. Before a state reaches the *mature* condition, its BPM is mapped exclusively to its internal state value, giving each unit an independent temporal identity. Once maturity is achieved, however, the BPM converges toward a unified global maximum.

This system enables a form of non-linear generative arrangement, in which individual behaviours diverge and evolve asynchronously but gradually synchronise as the ecosystem

matures. The interaction between local variation and eventual global coherence becomes a key compositional feature of the work.

Visual Engine

The visual layer of the project is built upon a set of nine plant images. I generated these using an AI-assisted workflow, ensuring that all images shared a consistent aesthetic style while remaining visually clean enough for further processing with jit.op. This allowed me to apply a series of stylisation operations that emphasised the internal structure of each leaf or petal. Through this process, I produced the final effect of luminous vein-like patterns, which form the foundation of the visual identity of the piece.

These processed images were then assembled using **jit.glue**, creating a composite layout that serves as the core visual field upon which the state-machine operates. The colour-modulation techniques described earlier determine how each element brightens, shifts hue, or fades according to its growth state.

In addition to the main collage layer, I implemented a **sparkling point system**, which introduces flickering highlights across the screen. This system is adapted from a patch developed in a previous project and was modified to integrate with the current state-machine and interaction logic.

Critical Reflection

A number of limitations became evident during the development of this prototype. First, the system lacks a deeper sense of spatial localisation in its audiovisual mapping. Although the interaction with light is effective, the absence of more nuanced spatial placement—whether through multi-channel audio, depth-aware visual behaviour, or spatialised mappings—reduces the overall immersive potential of the piece.

In terms of the compositional engine, the generative mechanism can at times feel somewhat limited or repetitive, especially during extended interaction. The current arrangement logic is functional but still leaves considerable room for expansion. Many parameters could benefit from finer-grained mapping, such as more diverse synthesis layers, additional effects modulation, or more sophisticated approaches to MIDI-scale quantisation and harmonic transitions.

These constraints are largely due to the substantial scope of the project within a relatively short production timeframe. Nevertheless, each of these aspects represents a clear direction for future iteration. Moving forward, I intend to further refine the mapping strategies,

extend the sound palette, and push this system toward a richer and more immersive generative ecology.

In addition to the main implementation, I conducted several experiments using **nn~** and timbre-modelling techniques, particularly the **RAVE** trained model. Unfortunately, the results were not sufficiently reliable or musically coherent to integrate into the final system. Direct inference as well as partial-processing experiments produced inconsistent timbral behaviour, and the large number of parameters available between the *encode* and *decode* stages made it difficult to establish a controlled or predictable mapping within the timeframe of this project.

Although I ultimately chose not to include this in technique this iteration, the process was valuable for understanding the challenges of real-time machine-learning-based sound transformation. This is an area I remain highly interested in, and I plan to explore it further in future.

Conclusion

Throughout this project, I have learned a great deal. I spent more time thinking about the conceptual foundations and aesthetic intentions behind building an interactive system, and I gained a clearer sense of how theoretical ideas can be meaningfully combined with practical implementation. I also experimented with several new technical approaches and integrated them with the skills I had already developed. This process was invaluable, both in shaping the project and in expanding my understanding of creative coding. Beyond the specific techniques used in this project, I also benefited from smaller but significant practices introduced along the way. For example, I have begun organising my workflow by packaging and reusing my own patch libraries, which has greatly improved my efficiency and allowed for more coherent project development. As I continue to work on new projects, I increasingly find that components from my earlier patches can be repurposed or adapted, creating a sense of iterative growth. This gives me the feeling of an evolving toolkit—one that will likely accelerate my creative process exponentially in future work.

Acknowledgement

I would like to express my gratitude to those whose teaching shaped this work for their guidance and feedback throughout its development.

I am also grateful to the developers and communities behind **Max/MSP, Arduino,** and the various external tools and libraries used in this work. Their publicly available resources and documentation allowed me to explore new technical directions and expand my practice.

Finally, I would like to acknowledge the artistic influence of **Brian Eno**, whose work on generative systems and ecological metaphors inspire my creative.

Third-Party Tools, Libraries, and Software Credits

Software & Libraries

MaxMSP 9 – Cycling ’74

Arduino IDE – Arduino.cc

Valhalla VintageVerb (VST) – Valhalla DSP

Scarred Piano (VST) — *Infinit Essentials* / Loot Audio

AI Images Generated with *ChatGPT — OpenAI Image Model*

Sound & Media Sources

Soundly:

Ambience, Forest, Birds, Wind In Trees 01 SND0432

Wind, Vegetation, Forest, Leaves 01 SND0247